

5ARE0 Data Analysis and Learning Methods

Assignment 2 - Detecting Emotions with ML Algorithms

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1 Introduction

In this project, emotion recognition is performed by training machine-learning models on image data.The main aim of the assignment is to compare the challenges involved in using different datasets and models for the same task.

2 FER2013 Baseline

2.1 Feature Extraction

Three types of feature representation were extracted from the 48×48 greyscale FER2013 images. All features were cached following their first extraction to enable fast re-run. HOG used 9 orientations, 8×8 pixels cells, and 2×2 block normalization (L2-Hys), resulting in a 900 dimensional vector. This method captures local edge and gradient structures in the vicinity of eyes, mouth corners, and eyebrows, being robust to some degree of illumination changes due to block normalization. Facial landmarks were extracted using dlib's 68 point shape predictor on the entire 48×48 patch, centered and normalized in max-abs way to 136 dimensions. Facial landmarks represent only geometric information about the face without any textural features. PCA normalized pixels to [0,1] and kept 253 dimensions that account for 95

2.2 Model Comparison

Three classifiers were tested on each feature type. For Logistic Regression (multinomial, with $C \in \{0.001, 0.01, 0.1, 1.0\}$), and for SVM (with RBF kernel, γ = scale, $C \in \{0.3, 1.0, 3.0, 10.0\}$), both classifiers employed StandardScaler. For Random Forest, 300 trees, `sqrt` number of features, entropy criterion, and `max_depth` $\in \{11, 20, \text{None}\}$ were used. Hyperparameters were tuned by means of validation macro-F1 (PublicTest), since FER2013 is heavily unbalanced — the Disgust class has only 436 train samples (1.5%) compared to 7,215 for Happy.

2.3 Performance Metrics

Table 1 reports all nine configurations on the PrivateTest split.

Table 1: FER2013 PrivateTest results for all nine configurations.

Feature	Classifier	Accuracy	Balanced acc.	Macro-F1
HOG	SVM	0.575	0.550	**0.568**
HOG	RF	0.489	0.426	0.450
HOG	LogReg	0.430	0.385	0.386
PCA	SVM	0.488	0.461	0.490
PCA	RF	0.406	0.352	0.376
PCA	LogReg	0.371	0.301	0.296
Landmarks	RF	0.489	0.435	0.452
Landmarks	SVM	0.486	0.423	0.438
Landmarks	LogReg	0.445	0.373	0.369

The optimal configuration chosen using validation macro-F1 is that of HOG+SVM with C=10 (validation macro-F1 = 0.562, test macro-F1 = 0.568). Similar scores between validation and test sets suggest that there is no overfitting to the validation set. Confusion matrices are in Appendix A.

2.4 Discussion of FER Results

Best classifier. SVM achieved the best macro-F1 results in all three feature types. In HOG and PCA, the RBF kernel's ability to produce complex nonlinear boundaries is superior to Logistic Regression's linear boundaries and

the axis-aligned splits of Random Forest. Logistic Regression had much lower F1s (HOG + LogReg: 0.386, compared to HOG + SVM: 0.568), indicating that the expression boundaries in HOG space have high nonlinearity. Random Forest worked fairly well for Landmarks (0.452), but poorly for HOG and PCA, where tree-based classifiers work bad with dense continuous features.

Best features. HOG outperformed PCA and Landmarks (max test macro-F1: HOG 0.568, PCA 0.490, Landmarks 0.452). HOG features represent the local patterns of textures around eyes and mouth which match muscle movements in expressions. PCA preserves the global structures of the pixels but is sensitive to the lighting variations in FER2013. Landmarks preserve only the geometric information and drop the textures. In the low resolution in-the-wild dataset where there is only approximate 48×48 alignment, 68-point dlib predictor works at the boundary of its reliable range.

Hardest emotions The hardest class was fear (average F1 across all nine models: 0.286), then angry (0.315) and sad (0.334). The easiest class was happy (0.627) due to the biggest number of training samples and distinctive pattern of wide smile. Surprise emotion was also recognized well (0.595). Even the best classifier (HOG+SVM) had the lowest scores for fear (0.458) and sad (0.433) emotions, which are often confused with each other and neutral, as those three emotions have similar geometries: slightly opened mouth and no wide smile. The most underrepresented emotion in FER2013 was disgust, scoring 0.622 in the best model despite its scarcity (1.5%).

Supporting evidence. The images misclassified with the highest confidence include those where the predicted class seems more correct than the ground truth label in FER2013, indicating annotation errors in crowd-sourced FER2013 labels. The confusion matrices show that the most common misclassification pairs in all models are fear–surprise and sad–neutral.

3 KDEF Study Section

Setup. The best FER2013 feature (HOG chosen by validation macro-F1=0.562) was used through the experiment. Only front images of KDEF were considered. Subject-dependent partition (56 train / 14 test subjects) ensures subject-independent testing — acquisition sessions of one person (A and B) are always in the same group. Hyper-parameters were optimized with 5 fold GroupKFold cross-validation by macro-F1 on the training subset grouped by subjects. 14 test subjects were used once for the final test score.

Results. Scores (with 95% bootstrap confidence interval) obtained with resampled subjects (1000 iterations) are shown in Table 2. The huge CI is due to the small size of the test set ($\tilde{1}$96 images from 14 subjects).

Table 2: KDEF test results with HOG features (95% bootstrap CI).

Classifier	CV macro-F1	Test accuracy	Test macro-F1	95% CI
LogReg	0.668	0.571	0.577	[0.498, 0.654]
SVM	0.640	0.602	0.609	[0.518, 0.690]
RF	0.619	0.633	0.619	[0.535, 0.716]

KDEF vs FER2013. KDEF scores were considerably higher than those for FER2013 (the best test macro-F1 for KDEF 0.619 vs 0.568 of FER2013, although same features and a much smaller train set of $\tilde{7}84$ images were used). KDEF expressions are easily distinguishable because posed and deliberately kept in a controlled environment with uniform background. The perfect class-balance (14 images per emotions per test subject) excludes possible class ignoring. Labels are assigned by the researchers rather than crowdsourcing. The huge bootstrap CI ([0.535, 0.716] for RF) results from having only 14 test subjects because one or two more or less successful affects the test score.

Classifier ranking. It changed drastically between the datasets. In the case of FER2013 SVM was ranked first and RF was ranked fourth with HOG features. For KDEF RF was ranked first (0.619) and LogReg was ranked second (0.577), SVM took the middle rank (0.609). Moreover, in all 9 configurations the collapse of the landmarks based classifiers occurred on KDEF (Landmarks + RF: rank 3 on FER $\rightarrow$ rank 9 on KDEF, macro-F1 0.452 $\rightarrow$ 0.265), while PCA + LogReg improved drastically (rank 9 on FER $\rightarrow$ rank 3 on KDEF, macro-F1 0.296 $\rightarrow$ 0.586). Small KDEF train set makes the use of the sophisticated models like SVM non-preferable; simple models such as RF and LogReg become relatively competitive when images are high quality.

Transferability of emotions. Happy and Surprise were correctly recognized on both sets, which is expected because of unique geometrical properties of emotions. The confusion of Sad and Neutral was observed on both of them confirming that this is not an issue associated only with FER but inherent to geometrically ambiguous emotions. Landmark-based models' collapse on KDEF is noteworthy – although KDEF contains better images, 20% margin crop leads to a smaller size of faces in 48×48 patch as compared to FER2013 and makes landmarding precision lower. Confusion matrices are shown in Appendix B.

4 Cross-Dataset Generalization

Setup and Adaptations. The best KDEF configuration (**HOG + LogReg**, selected by subject-grouped CV macro-F1 of 0.668) was re-fitted on 784 frontal KDEF images and evaluated directly on the 3,589 FER2013 test images with no fine-tuning. To run the model, we rescaled KDEF floats to 8-bit pixels, mapped KDEF emotion codes to FER2013 indices, and ensured no FER2013 data (including PCA standardization) leaked into the KDEF training pipeline. Face detection succeeded for all 980 frontal images. Because FER2013 is imbalanced (Happy has 879 images, Disgust only 55), we report macro-F1 alongside accuracy.**Results and Domain Shift.** As shown in Table 3, The model performs reasonably in-domain (macro-F1 0.577) but collapses on FER2013 (macro-F1 0.167, a loss of 0.41). The confusion matrices (Figure 2) show where the transfer breaks down: Neutral is practically never predicted (0.00 F1), Happy collapses from 0.91 to 0.28 F1, and Disgust drops to 0.01 F1.

	Accuracy	Balanced acc.	Macro-F1
KDEF test (in-domain)	0.571	0.571	0.577
FER2013 test (cross)	0.210	0.199	0.167
Chance (7 classes)	0.143	0.143	≈0.14

Table 3: HOG + LogReg trained on KDEF, evaluated in-domain and on FER2013.

This performance drop is caused by domain shift. **Covariate shift** is severe: KDEF contains studio photos with controlled lighting, whereas FER2013 contains low-resolution web images. A domain classifier confirms this, separating the datasets with 99.5% accuracy on HOG features, 99.4% on PCA, and 88.1% on landmarks. Furthermore, KDEF images are darker (brightness 83 vs 129) and have uniform contrast, while FER2013 is varied. **Label shift** (KDEF is balanced, FER2013 is not) and **concept shift** (KDEF is posed, FER2013 is spontaneous) further degrade performance. **Feature Transferability.** Table 4 compares the feature types. **Facial landmarks transfer best.** They are the weakest in-domain (0.302 on KDEF) but retain 69% of their score on FER2013, giving the best cross-dataset results (Landmarks + LogReg: 0.230 macro-F1). HOG and PCA are the best in-domain but keep only 20% of their score. This aligns with the domain classifier: geometry ignores texture and intensity, making it far less dataset-specific.

Feature	KDEF macro-F1	FER2013 macro-F1	Share retained
Landmarks	0.302	0.209	69%
PCA (pixels)	0.557	0.124	22%
HOG	0.602	0.126	21%

Table 4: Mean macro-F1 over LogReg, SVM, and RF, and the share of in-domain score retained.

5 Explainable Fuzzy Classifier

A fuzzy rule-based classifier was built on the frontal KDEF images (784 training / 196 test) using dlib to extract 68 landmarks. Six geometric features were computed, normalized by inter-ocular distance: mouth opening, mouth width, eye opening, eyebrow height, smile ratio, and brow gap. These features were chosen because they map to human-interpretable muscle movements (e.g., widening the mouth for smiling).

5.1 Geometric Feature Design

Appendix Figure 6 illustrates how the 68 dlib landmarks map to the six chosen geometric features on a frontal KDEF face. Figure 5 visualises the per-emotion distributions of these features across the training set. This reveals clear separability for some emotions: 'happy' has a distinctively high smile value (0.090) and widest mouth (0.789), while 'surprise' peaks in mouth opening (0.142) and eye opening (0.404). Conversely, 'sad' and 'neutral' show heavily overlapping distributions across all features, foreshadowing their poor classification results.

5.2 Fuzzy Rule Construction

Each feature is described by fuzzy terms (*low*, *medium*, *high*, *very high*) with breakpoints at training percentiles. The rule base is shown in Table 5. These rules were designed by inspecting the box plots and refined using the training confusion matrix.

Emotion	Rule
Happy	IF smile very high $\wedge$ mouth width very high
Surprise	IF mouth opening high $\wedge$ eye opening high $\wedge$ brow high $\wedge$ mouth width low
Afraid	IF eye opening high $\wedge$ mouth opening medium $\wedge$ mouth width medium
Angry	IF eyebrow height low $\wedge$ brow gap low $\wedge$ mouth opening low
Disgust	IF eye opening low $\wedge$ mouth opening medium
Sad	IF mouth opening low $\wedge$ eye opening medium
Neutral	IF mouth opening low $\wedge$ eye opening high

Table 5: Fuzzy rule base.

5.3 Interpretability Analysis

The classifier performs far above chance using only six numbers per face (Table 6). The test score (0.566) is slightly higher than the training score (0.529); with 196 test images, the sampling error alone is about ± 3.5 percentage points, so the difference is not evidence of better generalisation.

	Accuracy	Balanced acc.	Macro-F1
Training (784)	0.529	0.529	0.534
Test (196, unseen)	0.566	0.566	0.557

Table 6: Fuzzy classifier performance (chance: 0.143).

The confusion matrices (Figure 4) show that the classifier recognizes 'happy' almost perfectly (27/28) but struggles to distinguish between 'sad', 'neutral', and 'afraid'. Recall on test: happy 0.96, surprise 0.75, disgust 0.61, neutral 0.57, sad 0.46, angry 0.36, afraid 0.25. Several rules align well with human intuition. The "happy" rule uses the highest smile value (0.090) and widest mouth (0.789), achieving 96% recall. The "surprise" rule captures the expected "O"-shape through the widest mouth, eyes, and highest brows (75% recall). The "angry" rule identifies lowered/furrowed brows (36% recall). However, some rules are less intuitive. The "disgust" rule relies on narrowed eyes rather than a nose wrinkle, causing confusion with "happy" and "angry." The "afraid" rule lacks distinctive features and is often misclassified as "surprise" or "neutral." Finally, "sad" and "neutral" are difficult to distinguish geometrically (e.g., 11 neutral images were predicted as sad).

Although highly interpretable, the fuzzy classifier has limitations. It uses only six manually selected geometric features, missing subtle expressions such as nose wrinkling, upper-lip raising, or inner-brow raising. Its fuzzy breakpoints depend on the training distribution, and its rules were designed for frontal, posed faces. Performance is also limited by the dlib landmark detector's reduced accuracy on low-resolution 48×48 FER2013

images. Overall, the approach highlights the trade-off between transparency and classification accuracy.

6 Creative Challenge

6.1 Motivation and Methods

The reference was the Task 1 RBF support vector machine (SVM), using 900 standardized HOG features, $C = 10$ and $\gamma = \texttt{scale}$. Features were extracted from 48×48 grayscale faces using the same preprocessing as Task 1. The objective was to improve validation macro-F1, which weights all seven expression classes equally despite unequal frequencies.

Four modifications were investigated. First, regularization tuning compared $C \in \{10/3, 10, 30\}$, selecting 10/3 by validation macro-F1. Second, balanced class weights, $w_c = n/(7n_c)$, increased penalties for errors on less frequent Training classes. Weighting was tested at both baseline and tuned settings to separate their contributions.

Third, Training-only ANOVA feature selection retained 50% of standardized HOG dimensions, reducing the representation from 900 to 450 features. This could remove uninformative dimensions, but univariate ranking ignores feature interactions.

Finally, facial geometry supplemented HOG. Six landmark distances described mouth opening and width, eye opening, and eyebrow–eye gaps, each normalized by inter-eye distance. ANOVA ranking on 784 frontal images from 56 KDEF Training identities selected mouth opening, mouth width and left eye opening. These were appended to HOG, producing 903 features. Separate standardization branches preceded an unweighted RBF SVM with $C = 10$. This tested whether explicit shape information complemented gradients, at the cost of landmark extraction.

6.2 Experimental Protocol

The official FER2013 partitions were preserved: 28,709 Training images for fitting, 3,589 PublicTest images for validation and 3,589 PrivateTest images for final evaluation. Task 5 candidates used full Training; the fitted Task 1 model remained the control. The random seed was 42.

Scalers and FER feature selection were fitted on Training only. Geometry ranking excluded KDEF test identities, and class weights used Training frequencies. No augmentation, label cleaning or evaluation-set resampling was performed.

All configurations were compared on the same validation images. The highest validation macro-F1 determined the exported model before final testing, with balanced accuracy and accuracy used for ties. PrivateTest did not change the selection. Repeated comparisons on one validation partition nevertheless risk validation overfitting; no repeated-run significance analysis was conducted.

6.3 Results and Discussion

Table 7 isolates each modification and the tuning–weighting combination. Geometry produced the largest validation gain: 1.43 percentage points in macro-F1, 1.31 in accuracy and 1.41 in balanced accuracy.

Tuning improved macro-F1 by 0.37 points, whereas weighting alone reduced it by 0.10 points. Combining weighting with tuning also performed worse than tuning alone. Feature

Table 7: PublicTest validation results ($n = 3{,}589$); metrics in percent.

Configuration	Accuracy	Balanced acc.	Macro-F1
Task 1 control	56.98	54.05	56.24
Tuning only	57.37	54.41	56.61
Weighting only	56.90	53.95	56.14
Tuning + weighting	57.09	54.26	56.40
Feature selection	55.31	52.98	55.21
HOG + geometry	**58.29**	**55.46**	**57.67**

selection reduced macro-F1 by 1.03 points, suggesting that this pruning rule discarded useful information. Geometry retained baseline classifier settings and was not combined with tuning or weighting.

Table 8: PrivateTest results ($n = 3{,}589$); metrics in percent.

Model	Accuracy	Balanced acc.	Macro-F1
Task 1 control	57.54	54.99	56.83
Selected HOG + geometry	57.76	54.99	56.73

The validation gain did not generalize consistently (Table 8): test accuracy increased by 0.22 points, macro-F1 decreased by 0.10 points, and balanced accuracy was almost unchanged. Test F1 improved for Happy (0.7543 to 0.7656) and Surprise (0.7277 to 0.7438), but decreased for the remaining classes, including Disgust (0.6222 to 0.6067) and Sad (0.4331 to 0.4262). Disgust had only 55 test examples. Confusion matrices appear in Figure 1.

The HOG-plus-geometry model remained selected because it maximized validation macro-F1. Its deployment does not imply proven test improvement; switching models after inspecting PrivateTest would compromise evaluation.

7 Real-Time Demo

7.1 Model Adaptation

The demo loaded the model from Section 6 with its saved preprocessing. OpenCV converted camera frames to grayscale and detected faces using a frontal-face Haar cascade with scale factor 1.3 and five minimum neighbours. Crops were clipped to frame boundaries and resized to 48×48 using bilinear interpolation.

The shared extractor reproduced the 900 HOG values and three landmark ratios. Saved standardization and the SVM produced one of seven expression labels, displayed above the face. No retraining was required.

Both live tests enabled five-prediction majority voting; ties favour the newest tied label. History resets after detection gaps, multiple faces, feature failures or bounding-box intersection-over-union below 0.5.

7.2 Performance in Real-Time

Two informal tests used camera index 0 at 640×480. The first processed 180 frames at 11.84 FPS, detecting a face in 179 frames. The repeat processed 300 frames at 14.94 FPS, detecting a face in 284. Neither recorded feature-extraction failures. Throughput includes capture, processing and display.

The participant reported mostly matching neutral, happy, surprised and sad labels in the repeat, and correctly recognized sadness in the first run. Without frame-level annotations, these observations do not establish accuracy. Voting reduced label switches from 25 to 13 across 281 continuous single-face transitions, demonstrating display stability rather than improved recognition.

7.3 Challenges and Fixes

Shared extraction and saved preprocessing prevented training–deployment inconsistencies. The landmark predictor was loaded once and reused. Detection gaps displayed a no-face message and cleared voting history; invalid crops or feature failures skipped predictions rather than retaining stale labels.

Voting reduced switching but may delay genuine expression changes. Accuracy and delay were not measured, and lighting, distance, pose, glasses and hardware were not systematically documented. Robustness across these conditions remains untested.

Overall, the model supported an interactive single-person demonstration with mostly matching labels reported informally. Reliable performance across users remains unestablished, and predicted facial expressions do not verify internal emotional states.

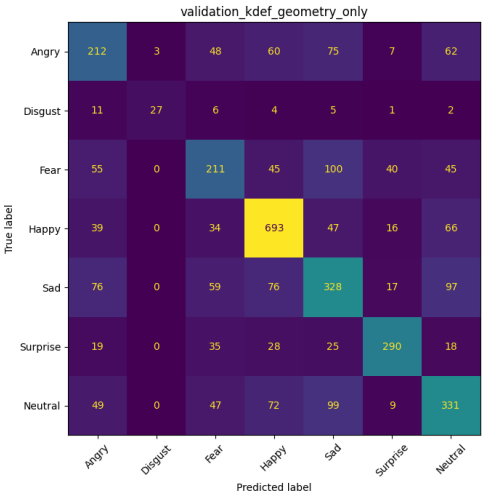
Use of AI

During the preparation of this work, We used ChatGPT and Claude in order to explain our code, create the structure and layout for our LaTeX report, and debug issues. The following parts of the assignment were affected/generated by AI tool usage: Methods, Code Explanation, LaTeX Formatting, and Discussion. After using this tool/service, Ewen Pouliquen, Mateusz Lotko, and Solomos Solomou evaluated the validity of the tool's outputs, including the sources that generative AI tools have used, and edited the content as needed. As a consequence, Ewen Pouliquen, Mateusz Lotko, and Solomos Solomou take full responsibility for the content of their work.

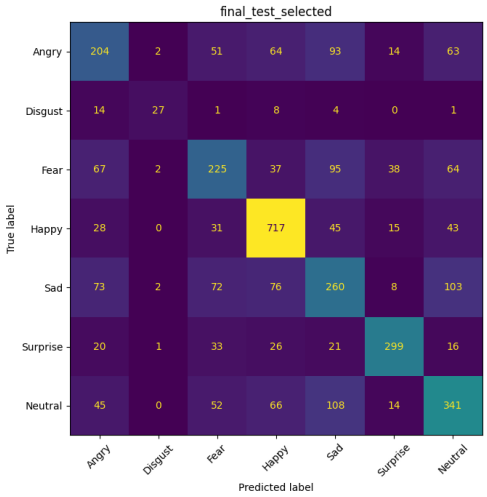
8 Conclusion

This project demonstrated that while models achieve strong in-domain accuracy on facial emotion recognition, they suffer severe degradation when transferred across datasets due to domain shift. Our experiments showed that KDEF-trained models collapse on FER2013 due to differences in acquisition and expression styles. However, geometric features (landmarks) proved far more robust to this shift than pixel-based features, retaining 69% of their performance. This motivated our interpretable fuzzy classifier, which achieved 56.6% accuracy and provided transparent, human-readable rules. While limited by a small set of handcrafted features, this work highlights the critical trade-off between model transparency and generalization.

A Creative Challenge Confusion Matrices



PublicTest validation ($n = 3{,}589$).



PrivateTest evaluation ($n = 3{,}589$).

Figure 1: Confusion matrices for the validation-selected HOG-plus-geometry model. Test results did not change the selected configuration.

B Cross-Dataset and Fuzzy Confusion Matrices

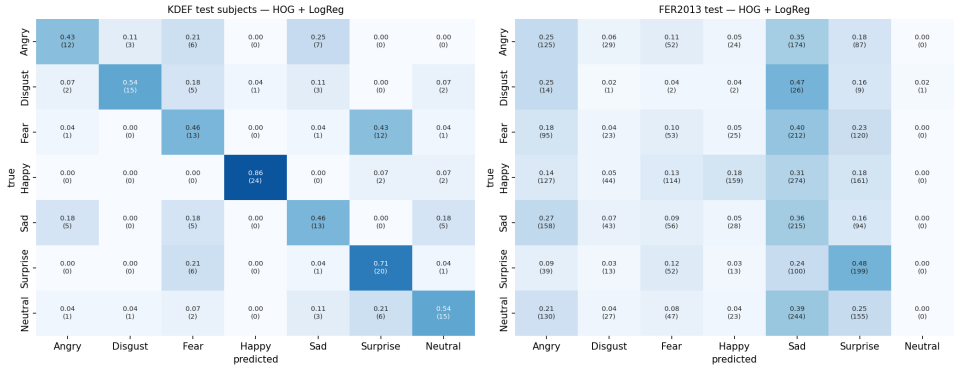


Figure 2: Row-normalised confusion matrices of HOG + LogReg on KDEF test subjects (left) and FER2013 test (right).

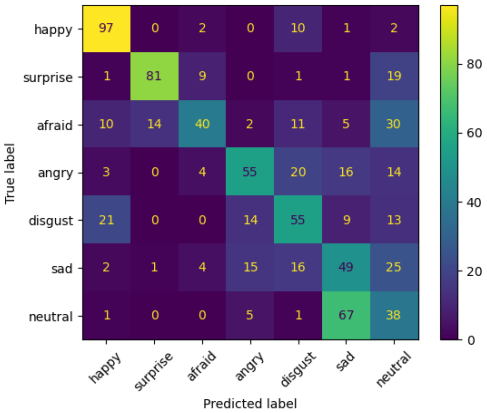


Figure 3: Confusion matrix – Fuzzy classifier (Training set)

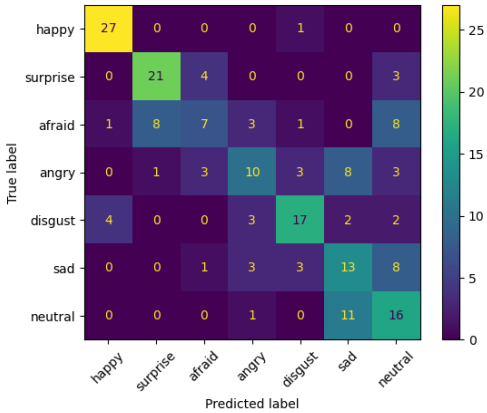


Figure 4: Confusion matrix – Fuzzy classifier (Test set)

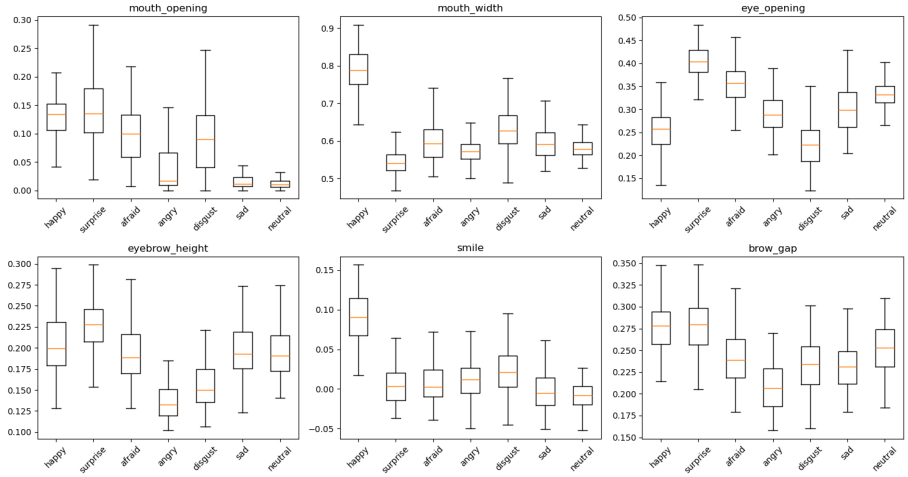


Figure 5: Feature distributions across emotions.

Feature	Definition
Mouth opening	Vertical gap between inner lips
Mouth width	Distance between mouth corners
Eye opening	Eye aspect ratio, mean of both eyes
Eyebrow height	Distance from brow to upper eyelid
Smile	Height of mouth corners relative to lip center
Brow gap	Distance between inner brow points

Table 9: Six handcrafted geometric features.

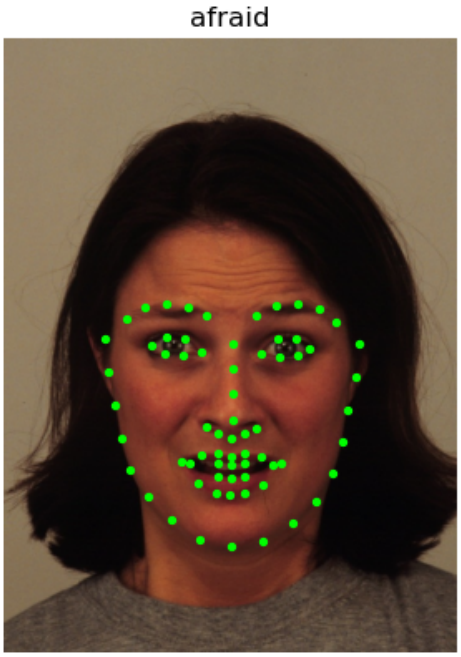


Figure 6: Facial landmarks used to compute features.

A FER2013 Confusion Matrices

B KDEF Confusion Matrices

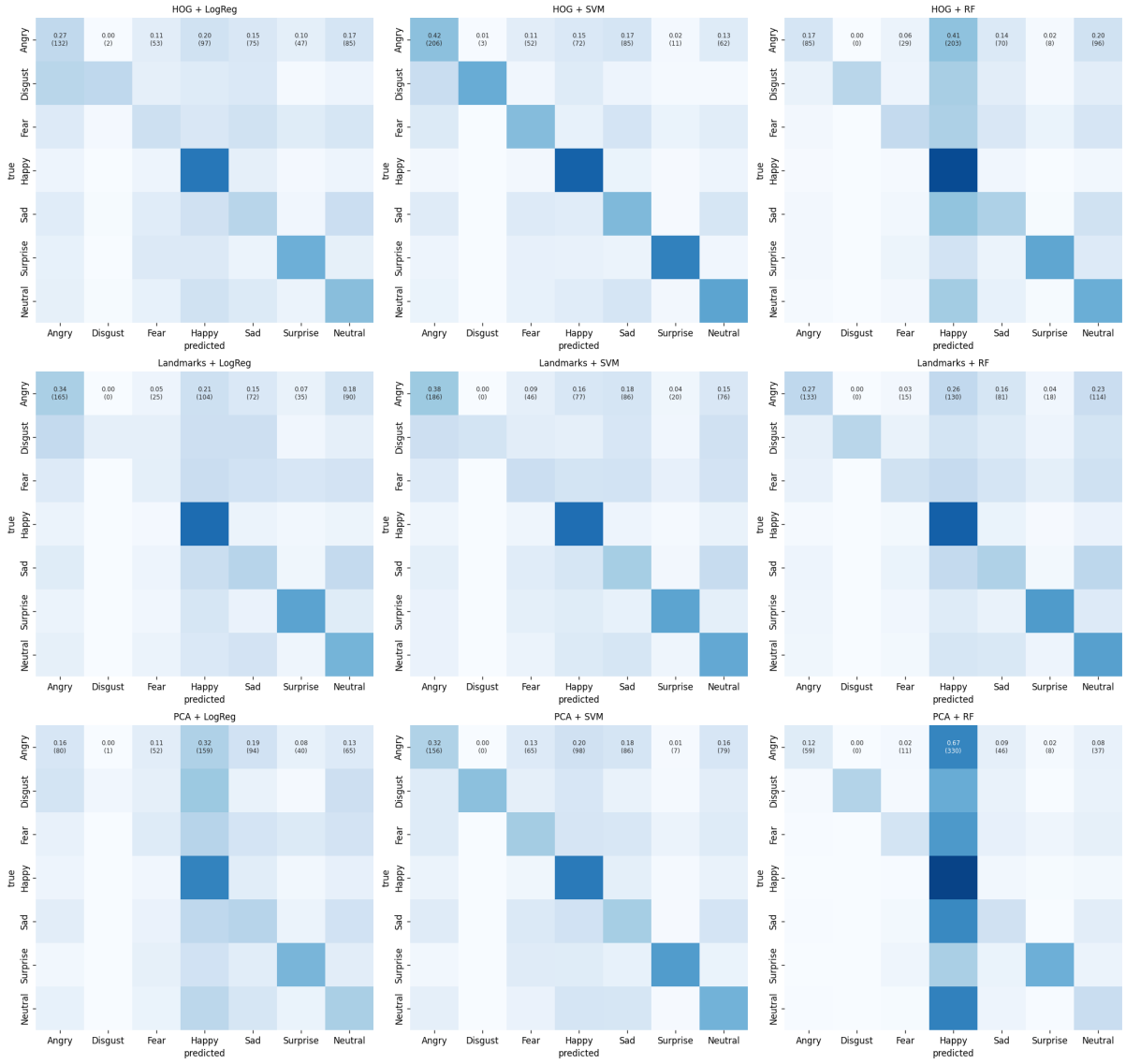


Figure 7: Row-normalised confusion matrices for all nine FER2013 configurations (PrivateTest).

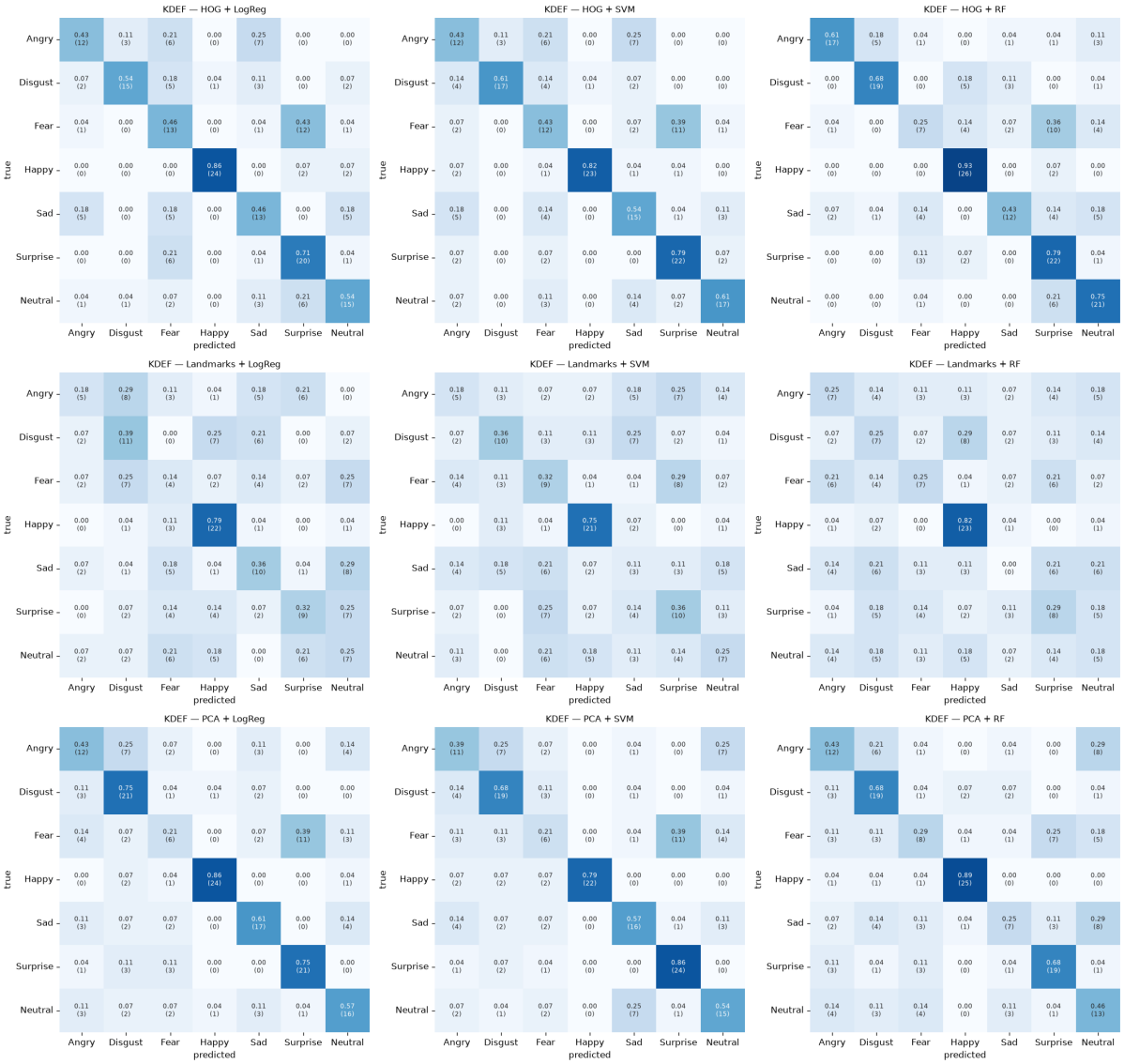


Figure 8: Row-normalised confusion matrices for all nine KDEF configurations (test subjects).